

■ Trapped Ambulance

How We Built an Automated Real-Life Green Corridor to Save Critical Lives

■ Hey there! Welcome to the simplest guide to this project.
Even if you have zero coding or electronics background, this document explains everything in simple, everyday terms: what problem we are solving, how our system turns traffic lights green automatically, what hardware and code we use, and why it saves lives during the *Golden Hour*.

■ 1. Why Are We Doing This? (The Real Problem)

Imagine someone you care about suffers a severe heart attack or a major highway accident. An ambulance picks them up and rushes to the hospital. But suddenly, **the ambulance gets stuck in bumper-to-bumper city traffic at a solid RED light!**

■ The 'Golden Hour' Reality:
In medical trauma care, the first 60 minutes after an acute emergency is called the **Golden Hour**. Medical research shows that every 60-second delay in critical care reduces patient survival by 7% to 10%! In crowded Indian cities, ambulances lose up to 30 to 45 minutes stuck at static red lights.

Why do traditional ambulances get stuck?

- **1. Sirens Are Ineffective:** Modern cars are soundproofed with air-conditioning and loud music. Drivers don't hear sirens until the ambulance is right behind them, by which time they are packed too tightly to move aside.
- **2. Traffic Lights are 'Blind':** Signals operate on rigid countdown clocks with zero awareness of approaching emergencies.
- **3. Blind Hospital Handover:** Emergency doctors have no idea of the incoming patient's condition until the ambulance physically arrives at the door.

■ 2. The Solution: Our Automated Green Corridor

Our project bridges the gap between the **Ambulance**, the **Traffic Lights**, and the **Hospital** in real time:

1. DETECT & TRACK ■ A GPS unit on the ambulance streams live location & speed 10 times every second to a central cloud server.	2. PREEMPT & CLEAR ■ When the ambulance enters 350m of a junction, the traffic light automatically switches to GREEN .	3. ALERT & PREPARE ■ The receiving hospital ER gets live patient ECG heartbeats and an exact countdown arrival clock.
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■ 3. How It Works Step-by-Step (The 4 Phases)

1■■ Phase 1: The Crisis	The ambulance carrying a cardiac patient is trapped behind 4 civilian cars at a solid RED light. Patient vitals are deteriorating.
2■■ Phase 2: Geofence Trigger	As the ambulance reaches 350 meters from the intersection, cloud GPS calculations fire an instant MQTT command. Nearby civilian drivers get a phone alert to pull over to the road shoulders, clearing the center lane!
3■■ Phase 3: Green Corridor	Our roadside microcontroller safely transitions cross-traffic with a 3-second Yellow light, and then locks the ambulance lane to solid GREEN . The ambulance zooms through with zero delay .
4■■ Phase 4: ER Handover	The hospital monitors the patient's live 60 FPS ECG heart graph and countdown ETA. Trauma surgeons and Operation Theater 03 are 100% prepared upon arrival — saving the life in the Golden Hour!

■ 4. The Hardware (Explained in Simple Terms)

Component	What is it?	What does it do in our project?
ESP32 Chip	A micro-computer smaller than a matchbox.	The Brain: Sits inside the traffic light box, receives the green corridor command, and flips the signal lights.
NEO-6M GPS	A GPS satellite receiver board.	The Eyes: Sits in the ambulance, tracking exact latitude, longitude, and speed 10 times a second.
4-Channel Relay	Electronic magnetic switches.	The Finger: Safely switches high-voltage 230V physical Red, Yellow, and Green traffic lamps ON/OFF.
Piezo Buzzer	A mini sound speaker.	The Voice: Plays the emergency siren audio during our prototype demonstration.

■ 5. What Code & Software Are We Using?

■ Frontend (What You See)	• index.html + Tailwind CSS: Sleek dark-mode glassmorphism mission control dashboard. • HTML5 Canvas: Animates the 2D traffic intersection, moving cars, and 60 FPS ECG heart waveform. • Web Serial API: Allows Google Chrome to directly control the physical ESP32 chip over USB!
■ Backend (The Engine)	• Python Flask (app.py): Lightweight web server handling emergency states, telemetry, and REST APIs. • MQTT Protocol: Sub-100ms ultra-fast messaging protocol sending preemption commands to traffic poles. • Arduino C++ Firmware: Code running inside the ESP32 chip managing the 30-second safe state machine.

■ 6. Why This is a Big Deal (The ■1,200 Miracle)

Comparison Metric	Traditional Smart City Overhauls	Our Trapped Ambulance System
Cost Per Junction	■25,00,000 (\$30,000)	Under ■1,200 (\$14.50)
Installation Time	6 to 12 months (digging roads)	15 minutes plug-and-play retrofit
Hospital Preparation	0 minutes (blind arrival)	10 to 15 min advance trauma prep
City Scalability	Only wealthy mega cities can afford	Scalable to every town & village globally!

■ The Big Pitch to Remember:

'For the cost of upgrading ONE single traditional smart junction, our system can retrofit over 2,000 intersections and turn an entire city into a life-saving green corridor. Because in the Golden Hour... saving seconds saves lives.'